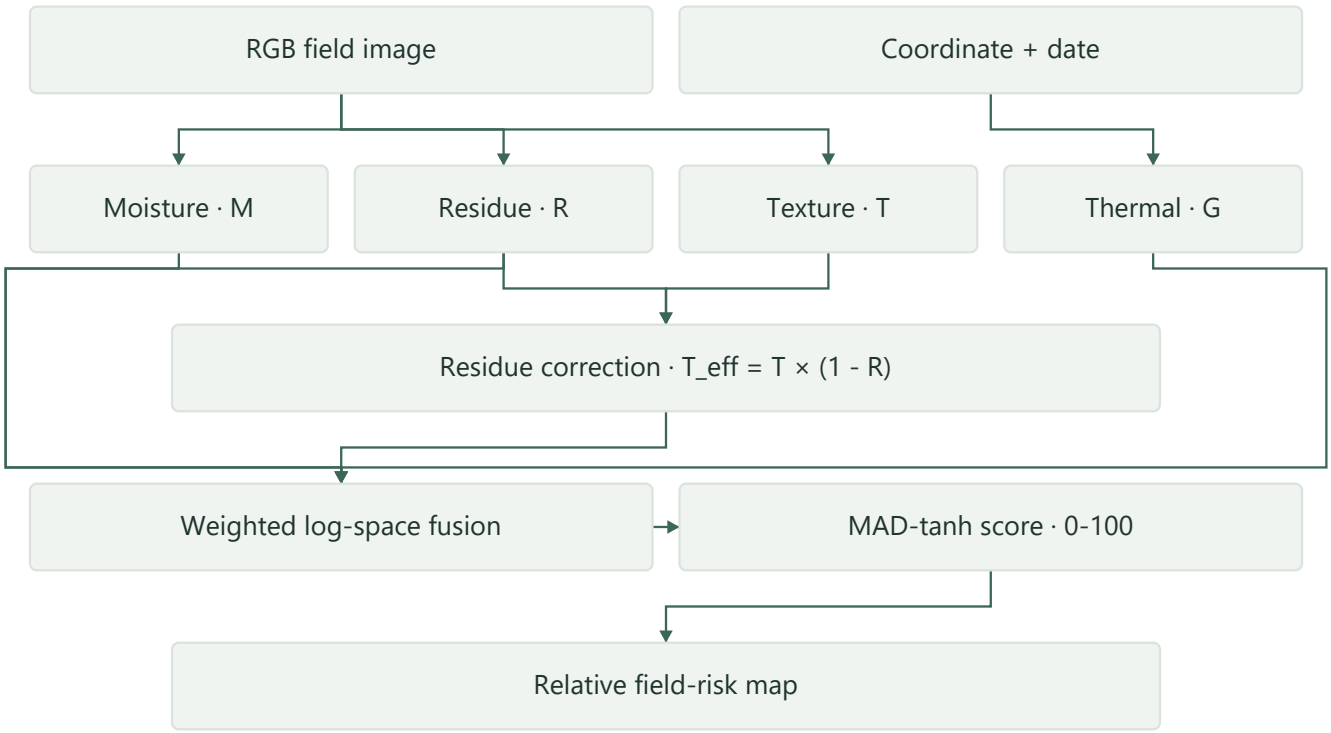
WERM

Weed Emergence Risk Mapping

A framework that combines field imagery and thermal accumulation to rank relative weed-emergence risk before visible emergence.

Architecture



Component overview derived from the supplied source; version-specific details may differ.

Component	Implementation	Role
Moisture	ResNet-18 regression	Image-derived wetness proxy
Residue	ResNet-18 + sigmoid	Fractional surface cover
Texture	32-level GLCM contrast	Surface-disturbance proxy
Thermal	7-day growing degree days	NASA POWER temperature input
Fusion	Weights 0.30 / 0.20 / 0.20 / 0.30	Relative score and A-E bins

Research prototype

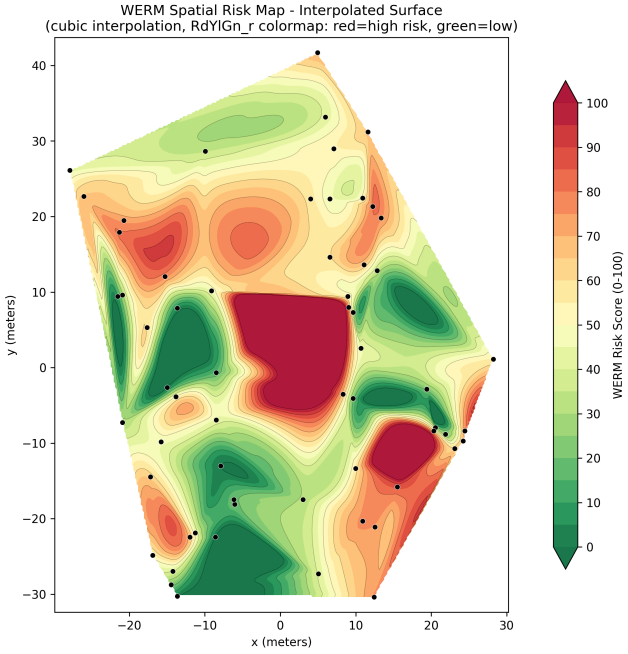
WERM | Selected evidence

Reported in the supplied manuscript and presentation

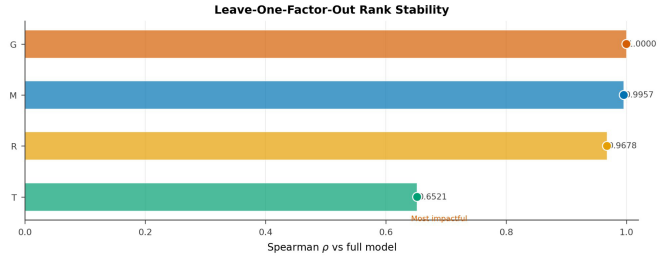
333
field images

61
spatial-grid images

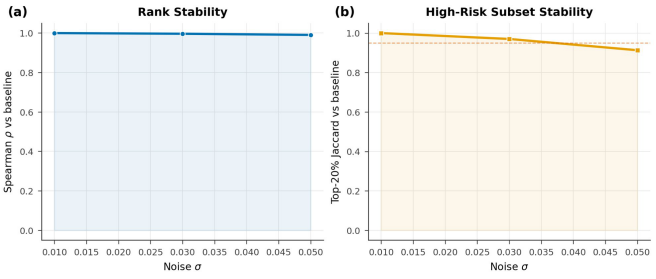
500
weight-ensemble draws



Relative risk surface · 61-image grid



Ablation: texture removal changes the ranking most.



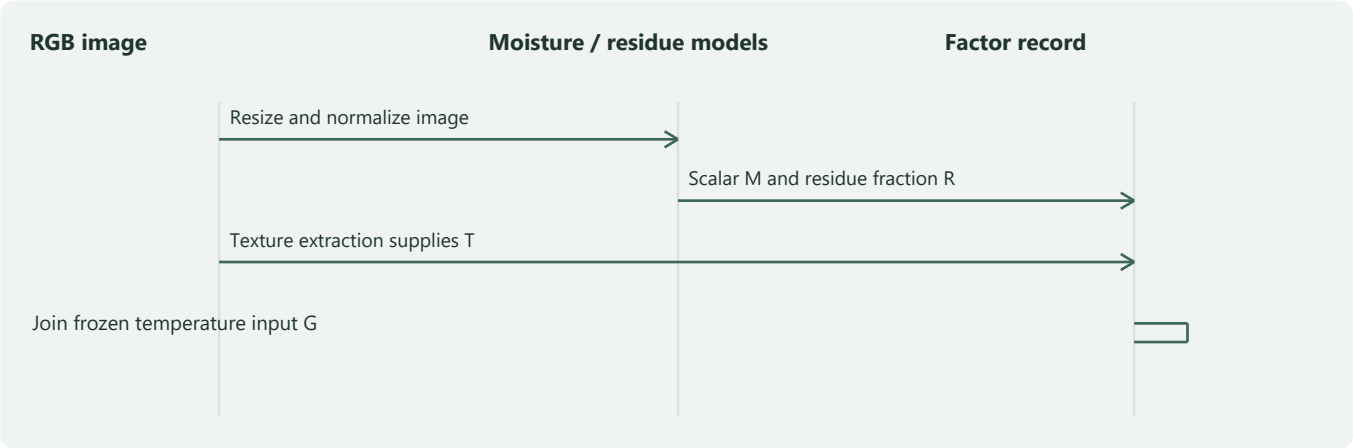
At 5% noise, reported rank correlation is 0.990.

Check	Reported result	Interpretation
Weight ensemble	42 / 333 images at high-risk inclusion ≥ 0.90	Across 500 sampled weights
Expert ranking	Geometric 0.652; additive 0.688	Overlapping reported confidence intervals
Fused-risk Moran's I	p = 0.777	Global autocorrelation not significant

Figures reproduced from the supplied WERM presentation. Relative ranking is not validated prospective emergence prediction.

From image to four factors

WERM separates learned image estimators from a deterministic fusion layer. This distinction matters: an image model can estimate a visual proxy well while the final index still needs validation against future emergence. The implementation contains multiple training and inference versions; this chapter describes their interfaces rather than treating every script as one frozen experiment.



Learned branches

The local moisture trainer is a ResNet-18 three-class model with cross-entropy, AdamW, stratified splitting, flips and colour jitter. The fusion path instead expects a regression-style head. The multitask definition contains classification and regression heads, but its matching training run is not established by the classification trainer. A checkpoint head is therefore part of the model identity, not an interchangeable file detail.

Residue and texture

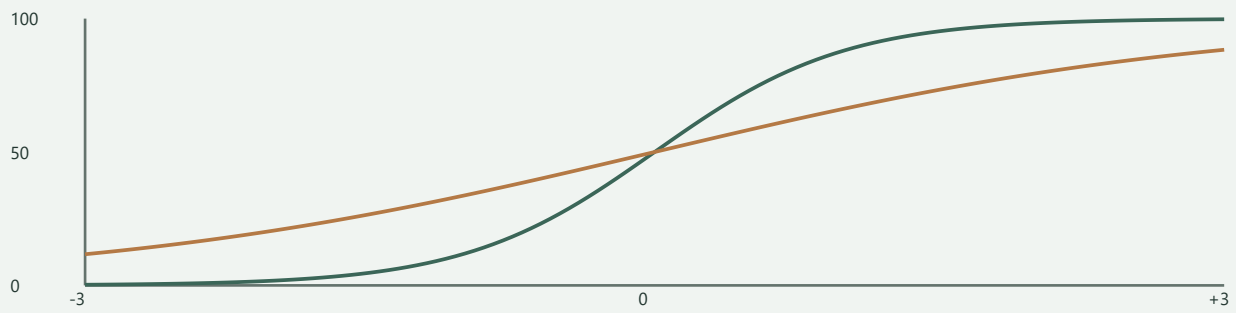
The residue trainer uses a sigmoid scalar head, CSV fraction labels and log-cosh loss, selecting checkpoints by validation MAE. Missing image rows can be dropped by its dataset loader. Texture alternatives include whole-image GLCM contrast and patch-based filtering. The patch-retention fraction describes retained coverage; it is not a calibrated probability of correctness.

Interface	Technical detail	Why it matters
M / R	Model head, input normalization, split and checkpoint	Defines what each scalar represents
T	GLCM levels, patch policy and normalization	Changes sensitivity to straw and edges
G	Latest seven valid temperature days; base 10 C in source	Valid days need not be consecutive calendar days

Source / interpretation: WERM/src/soil_moisture; WERM/src/residue_cover; src/tillage_module.py; src/temperature_gdd.py; moisture_multitask.py. Source defaults are not asserted as the final paper run.

Fusion mathematics & calibration

The geometric construction makes the assumptions inspectable. Let M denote moisture, G normalized thermal accumulation, R residue fraction, U = 1 - R, and T the normalized texture proxy. Damping defines T_eff = T * U^alpha before the four factors enter weighted logarithms. Values are floored by epsilon to make logarithms finite.



Standardized log-risk: -3 to +3. Green: engine. Ochre: integrated scaling.

Weighted log-risk

L = 0.30 ln(M) + 0.20 ln(G) + 0.20 ln(U) + 0.30 ln(T_eff), with epsilon flooring applied by the implementation. Before floors bind, this simplifies to 0.30 ln(M) + 0.20 ln(G) + (0.20 + 0.30 alpha) ln(U) + 0.30 ln(T). At alpha = 1 the uncovered-residue exponent is 0.50. Damping and the separate residue term consequently share information.

Why two score mappings matter

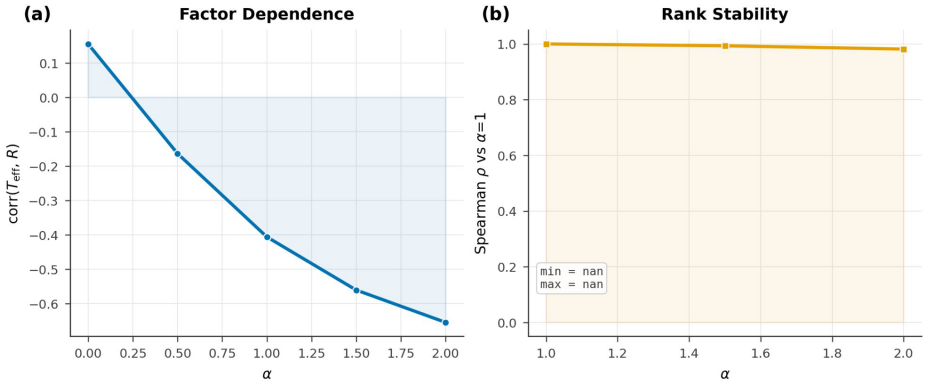
The reusable engine maps x = (L - median(L))/MAD(L) to 50(1 + tanh(x)). The integrated final script uses a wider scale, approximately 50(1 + tanh(x/(2*1.4826))); it also handles epsilon and near-zero spread differently. Both transforms are monotone for positive spread, but equal numeric thresholds can classify the same image differently. The plot isolates this scaling difference, not empirical performance.

Property	Consequence
Batch-relative scale	A changed comparison batch changes median/MAD and potentially the score of the same image.
Constant thermal term	A shared positive G adds a constant to L; refitting median/MAD removes that shift.
Index vs probability	A bounded 0-100 number does not imply a calibrated probability of emergence.

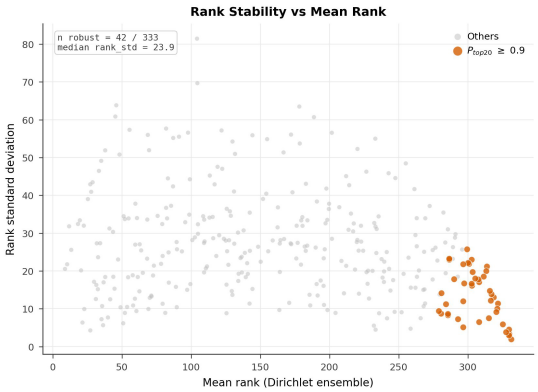
Source / interpretation: WERM/src/fusion_engine.py; fusion_final.py. Curves calculated from the two source formulas using standardized illustrative inputs.

Damping & weight uncertainty

These figures expose different questions: how residue alters a texture signal, and how much the selected high-risk set changes when weights vary. They are reproduced from the supplied presentation rather than redrawn as new experimental evidence.



Residue damping: supplied presentation, slide 8.



Weight ensemble: supplied presentation, slide 15.

Sensitivity is conditional

The ensemble uses 500 sampled Dirichlet weight vectors. The presentation reports 42 of 333 images with high-risk inclusion at least 0.90 under that sampling design. This is a frequency across sampled assumptions, not a prediction interval for weed counts or a guarantee over every possible weight vector.

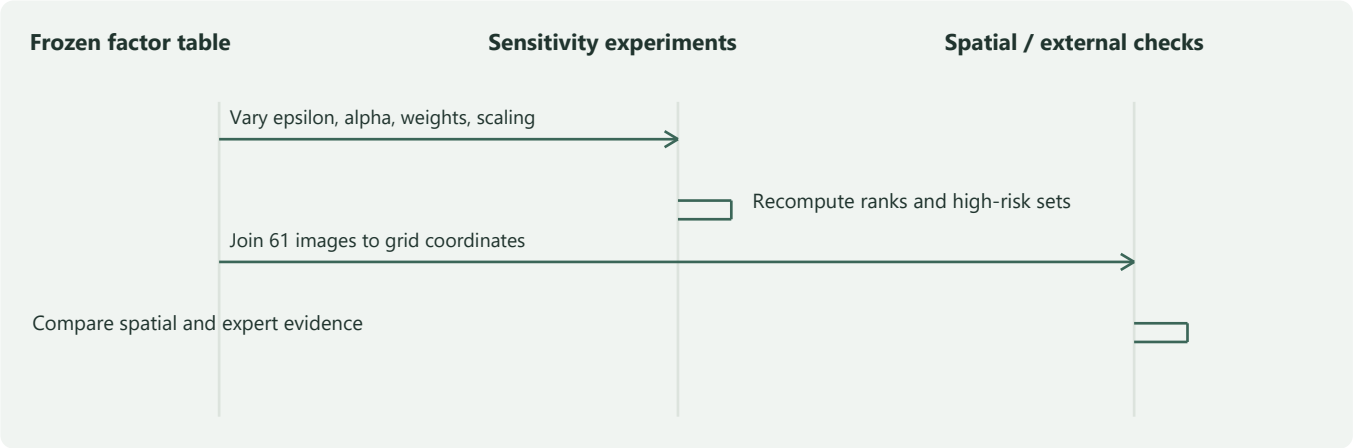
Interpreting variance

The presentation reports texture as 94.5% of the sum of component variances. The variance of a sum also includes pairwise covariance; these component shares are not independent causal effects. Residue-damped texture additionally contains the uncovered-residue factor, so a clean partition into four biological causes would overstate what this arithmetic establishes.

Source / interpretation: Supplied WERM presentation, slides 8, 12 and 15; local fusion/variance experiment scripts. Figures retain their original labels.

Evaluation design & spatial limits

The 333-image factor analysis and the 61-image spatial grid answer different questions. Rank agreement tests stability of ordering; grid diagnostics test spatial structure within one field layout. Neither alone measures prospective emergence prediction. A technical account needs to retain these units of analysis.



Observed results and interpretation

The supplied manuscript reports rank correlation 0.652 after texture removal and 0.990 at 5% input noise. Expert agreement is 0.652 for geometric fusion and 0.688 for additive fusion, with overlapping reported confidence intervals. These values do not establish a geometric-fusion advantage. The fused-risk Moran test has $p = 0.777$; significance in individual factors must not be transferred to the fused score.

Reproduction and future validation

Archive the exact 333-row factors, the 61-row coordinate join, model hashes, normalization settings, temperature response and random seeds with each figure. The alpha sweep should compute its alpha=1 reference before all comparisons; the inspected ordering leaves early comparisons missing. A subsequent validation should pair fixed pre-emergence scores with independently recorded counts and keep field-level development and evaluation separated. This is a proposed next test, not a completed result.

Diagnostic	Question it answers	Limit
Ablation / noise	How stable is the ranking?	Not prediction accuracy
Moran / variogram	How is variation arranged in space?	Range fits at the maximum lag can be censored
Expert agreement	Does ordering agree with annotations?	Requires annotation provenance and uncertainty

Source / interpretation: Supplied manuscript Tables 3 and 5; supplied presentation expert-agreement and spatial slides; WERM experiment scripts and existing deterministic verification.

Evidence & source

Implementation, supporting records, and the boundary of the claim

Area	Evidence	Reference
Primary image study	333 images from two wheat fields	Manuscript Section III.B
Ablation	Removing texture: rank correlation 0.652	Manuscript Table 3
Input perturbation	5% noise: rank correlation 0.990	Manuscript Table 3
Weight uncertainty	42 images with high-risk inclusion ≥ 0.90 across 500 sampled weights	Presentation slide 15
Spatial check	Fused-risk Moran's I: p = 0.777; not significant	Manuscript Table 5

Selected implementation files

File / record	Path / reference
fusion_engine.py	WERM/src/fusion_engine.py
tillage_module.py	WERM/src/tillage_module.py
temperature_gdd.py	WERM/src/temperature_gdd.py
experiment_runner.py	WERM/src/experiment_runner.py

Selected reusable modules. Full experiment notebooks, checkpoints, and frozen input records will be linked after the Drive archive is reconciled.

Evidence boundary

Relative ranking is the demonstrated output. Prediction against prospective weed-emergence counts remains unestablished. Scores are batch-relative, and sampled-weight stability is not a guarantee across every possible weighting.

Technical note

Residue damping changes the effective uncovered-residue exponent to 0.50 at $\alpha = 1$. The presentation clarifies that 94.5% refers to a share of summed component variances, not an independent causal attribution of total variance.